

Supplementary Materials *for*

**“A Precise and Real-time Eye-Tracking System Powered by Deep
Learning on Smartphones”**

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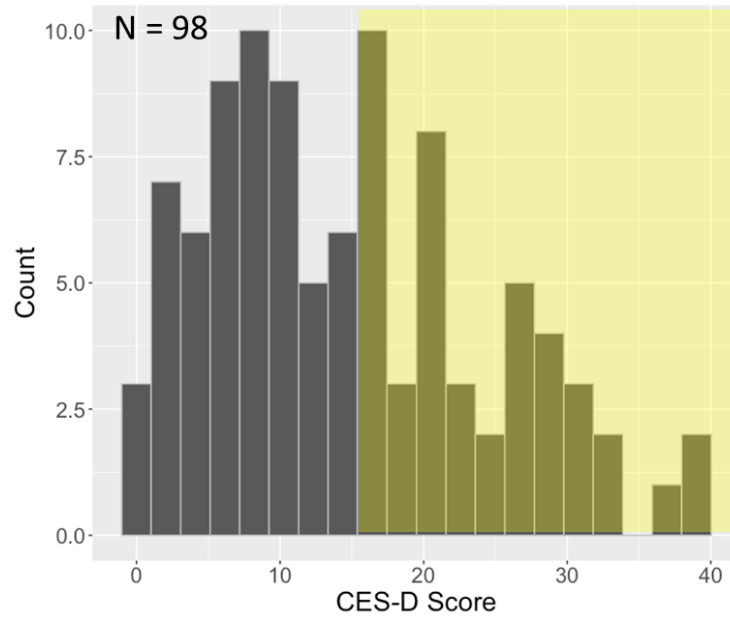


Figure S1. Distribution of the CES-D scores. The yellow area represents samples with CES-D scores equal to or greater than 16.

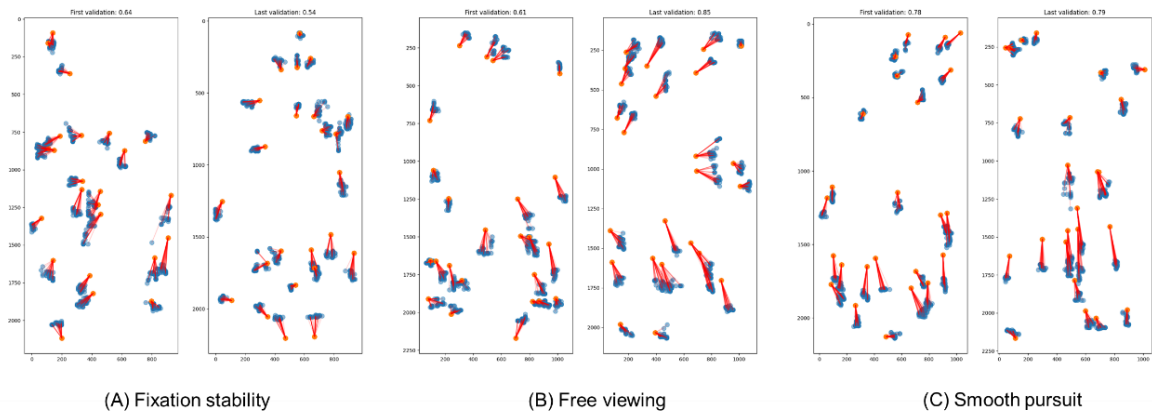


Figure S2. Gaze error in the pre- and post-task evaluation from a representative participant. The red dots represent the ground truth gaze of points, while the blue dots indicate the predicted gaze of points. The red bars represent the gaze error (Euclidean offset).

Table S1. Features extracted from the fixation stability task.

Feature name	Description
<i>fs sac cnt</i>	Saccade count
<i>fs sac dur mean</i>	Mean saccade duration
<i>fs sac dur std</i>	The STD of saccade durations
<i>fs sac amp mean</i>	Mean saccade amplitude
<i>fs sac amp std</i>	The STD of saccade amplitudes
<i>fs sac vel mean</i>	Mean saccade velocity
<i>fs sac vel std</i>	The STD of saccade velocities
<i>fs fix cnt</i>	Fixation count
<i>fs fix dur mean</i>	Mean fixation durations
<i>fs fix dur std</i>	The STD of fixation durations
<i>fs x mean</i>	Mean gaze position in the horizontal direction
<i>fs x std</i>	Variation in gaze position in the horizontal direction
<i>fs y mean</i>	Mean gaze position in in the vertical direction
<i>fs y std</i>	Variation in gaze position in the vertical direction

Table S2. Features extracted from the smooth pursuit task.

Feature name	Description
<i>sp sac cnt</i>	Saccade count
<i>sp sac dur mean</i>	Mean saccade duration
<i>sp sac dur std</i>	The STD of saccade durations
<i>sp sac amp mean</i>	Mean saccade amplitude
<i>sp sac amp std</i>	The STD of saccade amplitudes
<i>sp sac vel mean</i>	Mean saccade velocity
<i>sp sac vel std</i>	The STD of saccade velocities
<i>sp fix cnt</i>	Fixation count
<i>sp fix dur mean</i>	Mean fixation durations
<i>sp fix dur std</i>	The STD of fixation durations
<i>sp x offset mean</i>	Mean of gaze offset from the pursuit target in the horizontal direction
<i>sp x offset std</i>	The STD of gaze offset from the pursuit target in the horizontal direction
<i>sp y offset mean</i>	Mean of gaze offset from the pursuit target in the vertical direction
<i>sp y offset std</i>	The STD of gaze offset from the pursuit target in the vertical direction

Table S3. Features based on the eye movement data from all three tasks.

Feature name	Description
<i>exp fix cnt</i>	Fixation count
<i>exp fix dur mean</i>	Mean fixation duration
<i>exp fix dur std</i>	The STD of fixation durations
<i>exp x mean</i>	The mean gaze position in the horizontal direction
<i>exp x std</i>	The STD of gaze position in the horizontal direction
<i>exp y mean</i>	The mean gaze position in the vertical direction
<i>exp y std</i>	The STD of gaze position in the vertical direction

